Cardiovascular System (CARD) · NPLEX-P1-CARD-05

CV-5 Cardiac Muscle Contraction

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-05-01-C01-BD1

Brain dump 1 · foundational retrieval: Draw the ventricular action potential

A competent response contains

· Phase 4 rest is about -85 to -90 mV, held near EK (about -95 mV with K+ of 140 mmol/L inside and 4 mmol/L outside) by the inward rectifier IK1.

· Phase 0 begins at a threshold of about -70 mV and reaches about +20 mV within 1 to 2 ms through fast Na+ channels (Nav1.5, gene SCN5A), carrying positive charge inward; the steepness of this upstroke sets conduction velocity.

· Phase 1 is a brief notch from Na+ channel inactivation plus transient outward K+ current (Ito), and corresponds to the J point.

· Phase 2 holds the membrane near 0 mV for about 0.2 s, with sustained inward Ca2+ current through L-type channels (Cav1.2) balanced by small outward K+ currents, and corresponds to the ST segment.

· Phase 3 is repolarization: L-type channels inactivate, the delayed rectifiers IKr (KCNH2, hERG) and IKs (KCNQ1) grow, and below about -40 mV IK1 unblocks and finishes the job; this is the T wave.

· Phase 0 is the QRS complex, phase 4 is the interval from the T wave to the next QRS, and the whole action potential of about 0.25 to 0.3 s corresponds approximately to the QT interval.

· Inward rectification means IK1 conducts well near the resting potential but very little once the cell is depolarized, because intracellular Mg2+ and polyamines plug the pore.

· That property gives working myocytes a stable resting potential and limits K+ loss during the long plateau.

· Current arrives from an already depolarized neighbor through gap junctions in the intercalated discs, so the ventricle behaves as a functional syncytium and no myocyte needs its own nerve terminal.

Distinctions that matter

· The Ca2+ current of phase 2 is what holds the plateau; the Na+ current of phase 0 only sets the upstroke and is already inactivating by phase 1.

· IK1 and the delayed rectifiers are different K+ currents with different jobs: IK1 sets the resting potential and finishes phase 3, while IKr and IKs do most of the work of phase 3.

Common misconceptions

· Drawing the plateau as electrical quiet. Two large currents are flowing and nearly cancelling, which is why small changes in either one shift the duration of the action potential.

· Saying working myocytes drift to threshold on their own. They have abundant IK1 and sit still until a neighbor depolarizes them.

Where it is taught: Section C01, opening paragraphs, the ventricular-ap diagram, and the table of phases of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C01-BD2

Brain dump 2 · integration and application: Refractoriness and a change in plasma potassium

A competent response contains

· The effective refractory period lasts about 0.25 to 0.3 s and ends near -50 to -60 mV; during it no stimulus of any strength produces a propagated action potential because too few fast Na+ channels have recovered from inactivation.

· The relative refractory period lasts about 0.05 s; a stronger than normal stimulus fires the cell, and the upstroke is slower and smaller.

· Inactivated Na+ channels cannot reset until the membrane repolarizes, which is why refractoriness lasts nearly as long as contraction.

· Skeletal muscle has a brief refractory period, so twitches summate into tetanus; the cardiac refractory period covers most of the twitch, so the ventricle must relax and refill before it can be excited again.

· The long refractory period also resists reentry, and that protection is undermined where repolarization is uneven, as in ischemia or long QT syndromes.

· With K+ of 8.0 mmol/L, EK = 61.5 x log(8/140), about -76 mV instead of about -95 mV.

· Because IK1 dominates resting conductance, the resting potential follows EK up to about -70 mV.

· At that potential many fast Na+ channels remain inactivated between beats, so phase 0 is slower and smaller, conduction slows, and the QRS widens.

· High extracellular K+ also increases IKr conductance, so phase 3 speeds up and T waves become peaked, and this usually appears before the QRS widens.

· If most Na+ channels are inactivated, conduction fails and ventricular fibrillation or asystole can follow.

· A low plasma K+ of 2.6 mmol/L reduces IKr and IK1, so phase 3 lengthens and afterdepolarizations are favored.

Distinctions that matter

· High and low plasma K+ move phase 3 in opposite directions: high K+ speeds repolarization, low K+ slows it, even though both disturb rhythm.

· Peaked T waves are a repolarization change and a widened QRS is a depolarization change, which is why they appear in that order.

Common misconceptions

· Expecting hyperkalemia to make cells more excitable because the membrane is depolarized. The depolarization sits in the range that inactivates Na+ channels, so the cell becomes harder to excite and conducts poorly.

· Explaining the lack of tetany by saying cardiac contraction is slow. It is the length of the refractory period relative to the twitch that prevents summation.

Where it is taught: Section C01, refractory period heading and the worked example on hyperkalemia of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the dyad and the path from voltage to force

A competent response contains

· T tubules carry the action potential into the cell at the Z lines, and where a T tubule meets one sarcoplasmic reticulum terminal cisterna it forms a dyad with a cleft of about 12 to 15 nm.

· Across that cleft, L-type Ca2+ channels (dihydropyridine receptors, Cav1.2) face clusters of RyR2, and the store inside the sarcoplasmic reticulum is buffered by calsequestrin.

· Diastolic cytosolic free Ca2+ is about 0.1 µmol/L, peak cytosolic Ca2+ in a beat is about 1 µmol/L, and extracellular ionized Ca2+ is about 1.1 to 1.3 mmol/L.

· In human ventricle the sarcoplasmic reticulum supplies about 70 percent of activator Ca2+.

· Calcium-induced calcium release: trigger Ca2+ entering the cleft during the plateau binds the cytosolic side of RyR2 and opens it, releasing a much larger amount of Ca2+ from the store.

· One RyR2 cluster releasing Ca2+ gives a Ca2+ spark; thousands of synchronized sparks make up the Ca2+ transient.

· Release is graded because each L-type channel controls only the cluster it faces, so a larger trigger or a fuller store releases more Ca2+.

· Release ends within tens of milliseconds, mainly because falling local sarcoplasmic reticulum Ca2+ lowers RyR2 open probability.

· Cross-bridge cycle: Ca2+ binds the single regulatory site of cardiac troponin C, troponin C grips troponin I, troponin I lets go of actin, tropomyosin rolls deeper into the groove and uncovers myosin-binding sites, a myosin head carrying ADP and Pi binds actin, Pi leaves in the power stroke, ADP leaves into the rigor state, a new ATP must bind before the head detaches, and hydrolysis of that ATP recocks the head.

· Atrial myocytes have few T tubules, so their release spreads inward from the surface, and adult human ventricle expresses mainly the slower, economical beta myosin heavy chain.

Distinctions that matter

· Trigger Ca2+ and activator Ca2+ are not the same pool: a small entry through L-type channels releases the much larger store that actually binds troponin C.

· Cardiac troponin C has one regulatory Ca2+ site while fast skeletal troponin C has two.

Common misconceptions

· Saying ATP powers the power stroke itself. The power stroke comes from Pi release, and ATP binding is what detaches the head, with hydrolysis recocking it.

· Treating release as all or nothing. It is graded by the size of the trigger and the loading of the store, which is exactly what the heart regulates.

Where it is taught: Section C02, first three paragraphs and the excitation-contraction coupling flow of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-BD2

Brain dump 2 · integration and application: Remove the trigger, then let the store leak

A competent response contains

· In a Ca2+ free solution cardiac contractions weaken within seconds to a minute and stop, because no trigger Ca2+ enters to open RyR2 and the store runs down as the Na+/Ca2+ exchanger keeps extruding Ca2+.

· The skeletal muscle keeps twitching, because Cav1.1 opens RyR1 through a mechanical link and needs no extracellular Ca2+ for a twitch.

· The structural difference is the mechanical coupling of the skeletal triad against the diffusional, Ca2+ dependent coupling of the cardiac dyad.

· Blocking cardiac L-type Ca2+ channels removes the trigger, so force falls promptly, while skeletal twitches are spared for the same reason the Ca2+ free bath spares them.

· A leaky RyR2, or loss of calsequestrin, lets the store escape in diastole under adrenergic stress.

· The Na+/Ca2+ exchanger extrudes that leaked Ca2+ while bringing in 3 Na+ for each Ca2+, so it carries net inward positive charge and is electrogenic.

· That net inward current produces delayed afterdepolarizations, which can reach threshold and start ventricular arrhythmia during exercise.

· The condition is catecholaminergic polymorphic ventricular tachycardia.

· The notes describe the same mechanism in glycoside toxicity and in failing myocardium.

Distinctions that matter

· Loss of the trigger weakens contraction, while loss of diastolic RyR2 closure causes arrhythmia; both are RyR2 problems with opposite presentations.

· The exchanger is the arrhythmogenic step, not the leak itself, because it converts a Ca2+ movement into a depolarizing current.

Common misconceptions

· Expecting a Ca2+ free solution to stop skeletal twitches quickly. Skeletal release does not need extracellular Ca2+, so the twitch persists.

· Calling the exchanger electroneutral. It moves 3 Na+ in for 1 Ca2+ out, so each cycle carries net positive charge inward.

Where it is taught: Section C02, paragraph on coupling without a mechanical link and the stop and predict box of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-RT01

Retrieval exercise: Rebuild excitation-contraction coupling

A competent response contains

· Step 1: the action potential spreads through gap junctions and down the T tubules.

· Step 2: L-type Ca2+ channels in the T tubule membrane open and trigger Ca2+ enters the dyadic cleft.

· Step 3: Ca2+ in the cleft binds the cytosolic side of RyR2 and opens it, which is calcium-induced calcium release.

· Step 4: cytosolic Ca2+ rises from about 0.1 to about 1 µmol/L.

· Step 5: Ca2+ binds troponin C, troponin I releases actin, and tropomyosin moves to uncover myosin-binding sites.

· Step 6: cross-bridges cycle using ATP and the sarcomere shortens.

· Relaxation reverses the sequence: SERCA2a and the Na+/Ca2+ exchanger lower cytosolic Ca2+ to about 0.1 µmol/L, Ca2+ leaves troponin C, tropomyosin covers the binding sites again, and each attached myosin head binds a fresh ATP in order to release actin.

Distinctions that matter

· Ca2+ entry and Ca2+ release are separate steps in the sequence, and only the second one supplies most of the Ca2+ that binds troponin C.

· Relaxation is not simply the absence of excitation: it needs ATP for SERCA2a and ATP at every myosin head.

Common misconceptions

· Writing that the action potential opens RyR2. Voltage opens the L-type channel, and Ca2+ opens RyR2.

· Leaving ATP out of relaxation. A head without a fresh ATP stays attached, which is what produces rigor.

Where it is taught: Section C02, excitation-contraction coupling flow, with relaxation in C03 of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-CP01

Comparison table: Skeletal against cardiac excitation-contraction coupling

A competent response contains

· Excitation: skeletal fibers are excited by acetylcholine at their own neuromuscular junction; cardiac myocytes are excited by current from neighbors through gap junctions.

· T tubule position and junction: skeletal at the A-I junction forming a triad; cardiac at the Z line forming a dyad.

· Coupling: Cav1.1 mechanically opens RyR1 in skeletal muscle; in cardiac muscle Ca2+ entering through Cav1.2 opens RyR2, which is calcium-induced calcium release.

· Extracellular Ca2+: not needed for a skeletal twitch, required for a cardiac contraction.

· Action potential duration: about 1 to 5 ms in skeletal muscle, about 250 to 300 ms in cardiac muscle.

· Ca2+ sensor: troponin C in both, with two regulatory sites in fast skeletal troponin C and one in cardiac troponin C.

· A Ca2+ free solution leaves skeletal twitches largely unchanged within a minute, while cardiac force weakens and stops.

· Smooth muscle differs from both: it uses calmodulin and myosin light chain kinase rather than troponin C.

Distinctions that matter

· Both striated muscles use troponin C, so the real difference is in how the release channel is opened, not in the Ca2+ sensor.

· The long cardiac action potential is what supplies Ca2+ throughout the plateau, which is why its duration belongs in this comparison.

Common misconceptions

· Pairing the triad with cardiac muscle. The triad is skeletal, at the A-I junction; the cardiac junction is a dyad at the Z line.

· Assuming a drug that blocks L-type channels weakens both muscles. Skeletal coupling is mechanical, so it is spared.

Where it is taught: Section C02, comparison table on excitation-contraction coupling of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C03-BD1

Brain dump 1 · foundational retrieval: Build the calcium removal table from memory

A competent response contains

· SERCA2a sits in the sarcoplasmic reticulum membrane, uses ATP at 2 Ca2+ per ATP, removes about 70 percent of activator Ca2+, and is regulated by phospholamban.

· The Na+/Ca2+ exchanger sits in the sarcolemma, is driven by the Na+ gradient at 3 Na+ in per Ca2+ out, removes about 25 to 30 percent, and is governed by intracellular Na+ and membrane potential.

· The plasma membrane Ca2+-ATPase and mitochondrial uptake together handle about 1 to 2 percent, the pump using ATP in the sarcolemma under calmodulin control and mitochondrial uptake across the inner mitochondrial membrane.

· Cytosolic Ca2+ falls from about 1 µmol/L back to about 0.1 µmol/L during relaxation.

· Relaxation is active on two counts: SERCA2a and the plasma membrane pump hydrolyse ATP to move Ca2+ against its gradient, and each attached myosin head needs a fresh ATP before it can release actin.

· SERCA2a speed largely sets the rate of early diastolic relaxation.

· Unphosphorylated phospholamban lowers the Ca2+ affinity of SERCA2a and so inhibits it.

· Phosphorylation of phospholamban releases that inhibition, so phosphorylation activates the pump.

· Protein kinase A and CaMKII are the two kinases that phosphorylate phospholamban.

· The exchanger is electrogenic and depends on the Na+ gradient maintained by the Na+/K+-ATPase, so it slows or reverses when cytosolic Na+ rises or the membrane stays depolarized.

Distinctions that matter

· SERCA2a returns Ca2+ to the store inside the cell while the exchanger removes Ca2+ from the cell altogether, which is why one sets relaxation speed and the other sets the Ca2+ balance across the sarcolemma.

· Phospholamban is an inhibitor that is switched off by phosphorylation, so more phosphorylation means faster pumping.

Common misconceptions

· Describing relaxation as passive recoil. It costs ATP at the pumps and at every myosin head.

· Reading phospholamban phosphorylation as inhibition of SERCA2a. It removes inhibition.

Where it is taught: Section C03, first paragraph and the table of Ca2+ removal pathways of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C03-BD2

Brain dump 2 · integration and application: Partial inhibition of the sodium pump, beat by beat

A competent response contains

· Step 1: partial inhibition of the Na+/K+-ATPase raises cytosolic Na+ by a few mmol/L.

· Step 2: the smaller Na+ gradient slows the Na+/Ca2+ exchanger, so less Ca2+ leaves the cell during each diastole.

· Step 3: SERCA2a pumps the retained Ca2+ into the sarcoplasmic reticulum.

· Step 4: over several beats the store builds until extrusion again matches entry.

· Step 5: each beat then releases more Ca2+ through RyR2, so the Ca2+ transient is larger and contraction is stronger.

· Step 6: the effect needs no receptor signaling at all, which is why it works on a denervated or beta blocked heart.

· The change takes several beats because it works by accumulating store content, not by altering a single beat's trigger.

· With excessive inhibition the store becomes overloaded, releases Ca2+ spontaneously in diastole, and the exchanger turns that release into net inward current and delayed afterdepolarizations.

· Ischemia raises cytosolic Na+ by the same arithmetic: the Na+/K+-ATPase slows as ATP falls and Na+/H+ exchange adds more Na+ while the cell is acidotic, so the exchanger fails and Ca2+ accumulates.

Distinctions that matter

· Loading the store raises force over several beats, while a larger trigger raises force on the next beat; both act on release but on different timescales.

· A stronger beat and an afterdepolarization come from the same overload, which is why the margin between the two is narrow.

Common misconceptions

· Saying the pump inhibitor acts on the contractile proteins. It changes Na+, then the exchanger, then store loading.

· Expecting a receptor or a second messenger to be involved. This route is entirely transport based.

Where it is taught: Section C03, worked example on inhibiting the Na+/K+-ATPase and the paragraph on ischemia of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C03-RT01

Retrieval exercise: Rebuild calcium cycling in one heartbeat

A competent response contains

· Step 1: L-type channels admit a small trigger Ca2+ influx across the sarcolemma.

· Step 2: RyR2 releases most activator Ca2+ from the sarcoplasmic reticulum, about 70 percent of the total in human ventricle.

· Step 3: Ca2+ binds troponin C and cross-bridges generate force.

· Step 4: SERCA2a returns most Ca2+ to the sarcoplasmic reticulum, about 70 percent of what was removed.

· Step 5: the Na+/Ca2+ exchanger exports an amount equal to what entered, about 25 to 30 percent of removal.

· Step 6: cytosolic Ca2+ returns to about 0.1 µmol/L and the muscle relaxes.

· Balance one: reuptake into the store balances release from it.

· Balance two: extrusion across the sarcolemma balances entry across it.

· If extrusion falls behind entry, store content rises and later beats are stronger, up to the point of overload and diastolic release.

· If reuptake falls behind release, the store empties, diastolic Ca2+ stays high, relaxation slows, and later beats are weaker.

Distinctions that matter

· The two balances are separate: one governs store content, the other governs total cell Ca2+, and a pump inhibitor acts on the second to change the first.

· Most Ca2+ moves in and out of the store each beat while only a small amount crosses the sarcolemma.

Common misconceptions

· Treating the trigger influx as the main source of activator Ca2+. It is small, and its job is to open RyR2.

· Assuming entry and extrusion must be equal at every instant. They balance over the cycle, which is what lets store content change.

Where it is taught: Section C03, Ca2+ cycling cycle and the table of removal pathways of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C03-PR01

Prediction exercise: Predict what happens when coronary flow stops

A competent response contains

· Creatine phosphate falls quickly and anaerobic glycolysis produces lactate and H+, with Pi accumulating as ATP and creatine phosphate are broken down.

· Ca2+ sensitivity of troponin C falls because of the H+, and force per cross-bridge falls because of the Pi, so force drops within about 60 s while ATP is still substantial.

· Cytosolic Na+ rises as the Na+/K+-ATPase slows and Na+/H+ exchange imports Na+ in exchange for H+, so the Na+ gradient that drives the Na+/Ca2+ exchanger is lost and extrusion fails.

· SERCA2a slows as ADP and Pi accumulate and ATP falls, so diastolic cytosolic Ca2+ stays high and cross-bridges stay attached.

· The ventricle becomes stiff and end-diastolic pressure rises.

· Relaxation is impaired first, before reduced wall motion, ECG changes, and angina, which is the order of the ischemic cascade.

· After about 20 to 40 minutes Ca2+ overload causes irreversible injury, and on reperfusion that overload drives hypercontraction and contraction band necrosis.

· Severe ATP depletion leaves heads attached without fresh ATP, giving rigor and ischemic contracture.

Distinctions that matter

· Force falls from acidosis and Pi rather than from running out of ATP, which is why it falls so early.

· Impaired relaxation and impaired contraction come from the same energy failure, but they appear in a fixed order.

Common misconceptions

· Assuming contractile failure means ATP has run out. At 60 s ATP is still substantial and the problem is H+ and Pi at the myofilaments.

· Expecting ECG changes to be the earliest sign. Impaired relaxation comes first in the cascade.

Where it is taught: Section C03, heading on why ischemia impairs contraction and relaxation and the flow from ATP loss to stiff, weak myocardium of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C04-BD1

Brain dump 1 · foundational retrieval: Draw the beta-1 cascade and its targets

A competent response contains

· Norepinephrine comes from sympathetic nerve terminals and epinephrine from the adrenal medulla; both act on beta-1 receptors, the predominant beta receptor of ventricular myocytes.

· The beta-1 receptor couples to Gs; the alpha subunit stimulates adenylyl cyclase, mainly types 5 and 6, and cAMP rises.

· cAMP frees the catalytic subunits of protein kinase A, and A-kinase anchoring proteins hold the kinase close to its targets.

· L-type Ca2+ channel: phosphorylation raises open probability and lengthens openings, giving a larger trigger and more store release, so contraction is stronger.

· Phospholamban: phosphorylation removes its inhibition of SERCA2a, so reuptake is faster and store loading is higher, giving faster relaxation and stronger beats.

· Troponin I: phosphorylation lowers the Ca2+ affinity of troponin C, so Ca2+ dissociates faster and relaxation is faster.

· Myosin-binding protein C: phosphorylation speeds cross-bridge kinetics, so force rises and falls faster.

· IKs channel (KCNQ1): phosphorylation gives a larger repolarizing K+ current, so the action potential shortens at high rates.

· The net result within seconds is a stronger, faster, shorter twitch with faster relaxation: positive inotropy and positive lusitropy.

· The signal is ended by phosphodiesterases, mainly PDE3 and PDE4, by phosphatases, and by receptor desensitization through G protein-coupled receptor kinase 2 and beta-arrestin.

Distinctions that matter

· Inotropy is the force and speed of contraction at a given sarcomere length; lusitropy is the rate of relaxation, and phospholamban phosphorylation serves both.

· The first three targets are the core of the response; myosin-binding protein C and IKs shape the timing rather than the size of the transient.

Common misconceptions

· Treating faster relaxation as a side effect. A faster heart rate shortens diastole more than systole, so filling depends on faster relaxation.

· Leaving out the ways the signal ends. Phosphodiesterases, phosphatases and desensitization are what make the response brief.

Where it is taught: Section C04, first two paragraphs, the beta-1 signaling flow, and the table of protein kinase A targets of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C04-BD2

Brain dump 2 · integration and application: Take one target out of the cascade

A competent response contains

· Heart rate still rises and the L-type Ca2+ current still increases, so contractility increases somewhat in the mutant animal.

· SERCA2a stays inhibited, so reuptake does not speed up and the store grows less than it should.

· The normal phospholamban response does two jobs: within a beat it makes cytosolic Ca2+ fall faster, and across successive beats it loads the store so later releases are larger.

· Relaxation therefore accelerates far less, with mainly the troponin I effect left, and filling suffers at fast heart rates because diastole is short and the ventricle relaxes slowly.

· In a normal heart troponin I phosphorylation alone would slightly lower force at a given Ca2+, but the larger Ca2+ transient more than compensates.

· Enhanced IKs keeps repolarization in step with the larger plateau Ca2+ current, so losing IKs means the action potential fails to shorten when sympathetic drive raises the plateau current, which is why exercise provokes arrhythmias in long QT type 1.

· In chronic heart failure the notes give four changes: beta-1 receptors are downregulated, G protein-coupled receptor kinase 2 is increased, SERCA2a is reduced, and phospholamban is left underphosphorylated.

· Consequences: weaker contraction from a smaller trigger and a smaller store, slower relaxation from reduced and inhibited SERCA2a, a blunted response to exercise from fewer and desensitized receptors, and more arrhythmias from the resulting Ca2+ handling.

· RyR2 phosphorylation by protein kinase A is also reported, but its importance is debated.

Distinctions that matter

· Within a beat and across beats are two different effects of the same phosphorylation, which is why losing it costs both lusitropy and inotropy.

· Receptor level changes and downstream protein changes both blunt the response in failure, and they are separate faults in the same pathway.

Common misconceptions

· Expecting no inotropic response at all in the mutant. The L-type channel and the rate response are intact.

· Reading troponin I phosphorylation as a weakening of the heart. The bigger transient outweighs it, and the gain is faster relaxation.

Where it is taught: Section C04, paragraph on phospholamban, the final paragraph on heart failure, and the stop and predict box of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C04-CA01

Clinical application: Breathlessness on exertion in a man with a dilated ventricle

A competent response contains

· Recognize: the failing pathway is beta-1 adrenergic signaling through Gs, cAMP and protein kinase A, shown by reduced beta-1 receptor density, increased G protein-coupled receptor kinase 2, reduced SERCA2a, and largely unphosphorylated phospholamban.

· Explain: fewer receptors and more G protein-coupled receptor kinase 2 mean less cAMP and less protein kinase A activity, so the L-type Ca2+ channel is phosphorylated less and the trigger is smaller; unphosphorylated phospholamban keeps SERCA2a inhibited and reduced SERCA2a protein lowers pumping further, so reuptake is slow, the store is underfilled, the Ca2+ transient is smaller and slower to fall, and troponin I stays unphosphorylated so Ca2+ leaves troponin C slowly.

· Differentiate: in low filling the Ca2+ transient is normal and the heart simply sits lower on a normal ventricular function curve, so force at any given sarcomere length is normal; here the transient is smaller and the whole curve is shifted downward, so force is low at every sarcomere length.

· Investigate: on the exercise test the failure of heart rate to rise much above resting and the flat stroke volume show chronotropic and inotropic responses to catecholamines are blunted despite elevated plasma norepinephrine; on the echocardiogram the dilated ventricle with slowed early diastolic filling shows the relaxation defect, and filling pressures that are adequate rule out simple underfilling.

· Decide: slowed early diastolic filling comes mainly from SERCA2a, which is both reduced in amount and left inhibited by unphosphorylated phospholamban, and once heart rate rises the shortened diastole leaves too little time for a slow pump to lower cytosolic Ca2+, so filling falls further.

· Justify: SERCA2a normally removes about 70 percent of activator Ca2+ and its speed largely sets early diastolic relaxation, so inhibition plus reduced protein slows that removal directly; the Ca2+ it fails to pump back leaves a smaller store, and because release is graded by store content, later beats release less Ca2+ and contract more weakly, which is how one pump defect produces both the stiff filling and the weak ejection.

Distinctions that matter

· A preload problem and an inotropic problem both lower stroke volume, and the Ca2+ transient is what separates them.

· Reduced SERCA2a protein and unphosphorylated phospholamban are two independent reasons for the same slow reuptake.

Common misconceptions

· Reading the high plasma norepinephrine as evidence that beta-1 signaling is intact. The drive is high precisely because the response is blunted.

· Treating slow relaxation as a passive stiffness problem only. Ca2+ removal is active, and its failure is what keeps cross-bridges attached in diastole.

Where it is taught: Section C04, final paragraph on chronic heart failure, with Ca2+ removal in C03 and curve shifts in C05 of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the length-tension curves

A competent response contains

· Thick filaments are about 1.6 µm and thin filaments about 1 µm long, so overlap is optimal at about 2.0 to 2.2 µm.

· Ventricular sarcomeres work from about 1.8 to 1.9 µm at low filling to about 2.2 µm at normal to high filling, which is the ascending limb.

· On the descending limb stretch reduces overlap until active force reaches zero near 3.6 µm in the classic skeletal muscle studies.

· On the ascending limb force is limited because thin filaments from opposite sides interfere with each other, and at very short lengths thick filaments butt against the Z lines.

· Length-dependent activation is the rise in myofilament Ca2+ sensitivity with stretch, which shifts the force versus Ca2+ curve to the left so the same Ca2+ transient produces more force.

· The three proposed mechanisms are titin strain positioning more myosin heads to bind actin, narrower filament spacing as the cell thins, and cooperative Ca2+ binding to troponin C.

· Over the working range force rises several times more than filament overlap alone predicts, which is why the cardiac curve is steep.

· Stiff cardiac titin isoforms and collagen make passive tension climb steeply above about 2.2 µm, so intact hearts rarely reach the descending limb.

· A slower additional rise in the Ca2+ transient develops over minutes after sustained stretch, but the beat-to-beat response comes from overlap and Ca2+ sensitivity.

Distinctions that matter

· Active and passive tension are separate curves: active tension comes from cross-bridges, passive tension from titin and collagen.

· Overlap and Ca2+ sensitivity are two separate contributions to the same steep ascending limb.

Common misconceptions

· Drawing the ventricle working near the peak of the curve. It works on the ascending limb, below the optimum.

· Explaining the steep cardiac curve by overlap alone. Overlap changes modestly over the working range and sensitivity supplies most of the rise.

Where it is taught: Section C05, first two paragraphs and the length-tension comparison table of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C05-BD2

Brain dump 2 · integration and application: Tell a preload change from an inotropic change

A competent response contains

· Skeletal muscle grades force by recruiting motor units according to the size principle and by summating twitches at higher firing rates.

· The heart cannot recruit because gap junctions make it a functional syncytium and every myocyte contracts on every beat, and it cannot summate because the refractory period lasts nearly as long as the twitch.

· Lever one is Ca2+ delivery: raised by beta-1 stimulation and by a higher rate, which loads the store in the positive force-frequency relationship, and lowered in failing hearts where that relationship is blunted.

· Lever two is myofilament Ca2+ sensitivity: raised by stretch, lowered by acidosis and by troponin I phosphorylation.

· Lever three is sarcomere length, set by filling.

· These levers work only because a normal Ca2+ transient does not saturate troponin C, so there is room for the same transient to produce more or less force.

· The first heart shows a preload effect: it moves along its existing ventricular function curve with little change in the Ca2+ transient, and force at a given sarcomere length is unchanged.

· The second shows an inotropic effect: the larger transient shifts the curve upward, so force rises at any sarcomere length, and this is what a beta-1 agonist produces.

· A dilated failing ventricle is weak from reduced contractility, higher wall stress, and mitral regurgitation, not from overstretch.

· Chronic volume overload adds sarcomeres in series, which keeps individual sarcomeres off the descending limb even as the chamber enlarges.

· A stiff ventricle from hypertrophy, fibrosis or ischemia needs a higher filling pressure to reach the same volume.

Distinctions that matter

· Moving along a curve and shifting a curve are different events, and the Ca2+ transient tells you which one happened.

· Chamber dilation and sarcomere overstretch are not the same thing, because sarcomeres are added in series.

Common misconceptions

· Explaining failure in a dilated heart as sarcomeres on the descending limb. Stiff titin and collagen, plus added sarcomeres, keep them off it.

· Saying the heart recruits more myocytes when it needs more force. Every myocyte already contracts on every beat.

Where it is taught: Section C05, third and fourth paragraphs and the stop and predict box of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C05-CP01

Comparison table: Length-tension and force grading in skeletal against cardiac muscle

A competent response contains

· Working sarcomere length: skeletal muscle operates near the optimum, set by its bone attachments; cardiac muscle operates on the ascending limb at about 1.8 to 2.2 µm.

· Basis of the active length-tension curve: filament overlap in skeletal muscle; overlap plus length-dependent Ca2+ sensitivity in cardiac muscle, which makes the curve steep.

· Passive tension at long lengths: low in skeletal muscle until well beyond the optimum; high in cardiac muscle above about 2.2 µm because of stiff titin isoforms and collagen.

· Recruitment and summation: available in skeletal muscle through motor units and tetanus; unavailable in cardiac muscle because all cells contract each beat and the refractory period is long.

· Ca2+ delivery and sensitivity: a minor role in skeletal force grading, the major levers in cardiac force grading.

· Starting length: not used physiologically in skeletal muscle; a major lever in the heart, which is the Frank-Starling mechanism.

· Skeletal muscle varies how many fibers are active and how often they fire, while cardiac muscle varies how strongly each myocyte contracts.

Distinctions that matter

· Both muscles have a length-tension relationship, but only the heart uses starting length as a physiological control.

· High passive tension is protective in the heart, because it is what keeps sarcomeres off the descending limb.

Common misconceptions

· Filling the recruitment row for the heart with a partial answer. The heart has no recruitment and no summation, and the reasons are different for each.

· Writing that cardiac muscle works at its optimal overlap. It works below the optimum, on the ascending limb.

Where it is taught: Section C05, length-tension and force grading comparison table of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C01-Q01

Board-style question: A widened QRS with a high plasma potassium

Correct answer: C. A less negative resting potential leaving fast Na+ channels inactivated

Why it is right: Hyperkalemia moves the resting potential to about -70 mV because IK1 holds the membrane near EK, and at that voltage most fast Na+ channels remain inactivated, so the phase 0 upstroke is slow and small, ventricular conduction velocity falls, and the QRS widens.

A is wrong: Increased conductance of the rapid delayed rectifier K+ current is a real effect of hyperkalemia, but it speeds phase 3 and produces the peaked T waves rather than widening the QRS.

B is wrong: Prolonged opening of L-type Ca2+ channels would lengthen the plateau and the ST segment and QT interval, not the duration of ventricular depolarization.

D is wrong: Reduced Ca2+ release through RyR2 would weaken contraction, and release from the sarcoplasmic reticulum generates no deflection on the surface tracing.

A competent response contains

· A plasma K+ of 8.1 mmol/L raises EK from about -95 mV to about -76 mV, and because IK1 dominates resting conductance the resting potential follows to about -70 mV.

· At that voltage many fast Na+ channels stay inactivated between beats, so phase 0 is slower and smaller, conduction slows, and the QRS widens.

· The peaked T waves come from the separate effect of high extracellular K+ on IKr, which speeds phase 3.

Distinctions that matter

· The T wave change is a repolarization effect and the QRS change is a depolarization effect, with different currents behind each.

Common misconceptions

· Expecting a depolarized resting potential to make the cell easier to excite. It sits in the range that inactivates Na+ channels.

Where it is taught: Section C01 of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C02-Q02

Board-style question: Two muscles in a calcium free bath

Correct answer: A. Cardiac release channels open only in response to Ca2+ entering through L-type channels, while the skeletal voltage sensor opens its release channel mechanically

Why it is right: Cardiac excitation-contraction coupling depends on trigger Ca2+ crossing the sarcolemma to open RyR2, so a Ca2+ free bath abolishes release within seconds to a minute, while the skeletal voltage sensor Cav1.1 opens RyR1 mechanically and keeps releasing the internal store.

B is wrong: Calmodulin and myosin light chain kinase belong to smooth muscle; both skeletal and cardiac muscle use troponin C as the Ca2+ sensor.

C is wrong: The cardiac sarcoplasmic reticulum holds a large store buffered by calsequestrin, and it supplies about 70 percent of activator Ca2+ in human ventricle.

D is wrong: The A-I junction and the triad are the skeletal arrangement; the cardiac T tubule meets the sarcoplasmic reticulum at the Z line in a dyad with a cleft of about 12 to 15 nm.

A competent response contains

· Cardiac coupling is Ca2+ dependent: trigger Ca2+ entering the dyadic cleft through L-type channels opens RyR2, so removing extracellular Ca2+ removes the trigger and the store also runs down through the Na+/Ca2+ exchanger.

· Skeletal coupling is mechanical: Cav1.1 in the triad opens RyR1 through a physical link, so a twitch needs no extracellular Ca2+.

Distinctions that matter

· Both muscles have a sarcoplasmic reticulum store and both use troponin C; the difference is only in how the release channel is opened.

Common misconceptions

· Assuming any Ca2+ dependent muscle must fail without extracellular Ca2+. Skeletal muscle depends on its internal store and a mechanical trigger.

Where it is taught: Section C02 of the CV-5 Cardiac Muscle Contraction notes.

NPLEX-P1-CARD-05-01-C05-Q03

Board-style question: Force doubles with a small change in overlap

Correct answer: D. Greater myofilament Ca2+ sensitivity at the longer sarcomere length

Why it is right: Length-dependent activation raises myofilament Ca2+ sensitivity when the sarcomere is stretched, through titin strain, narrower filament spacing and cooperative Ca2+ binding to troponin C, so the same peak Ca2+ produces several times more force than the change in overlap alone would give.

A is wrong: Recruitment does not exist in the heart, because gap junctions make it a functional syncytium and every myocyte contracts on every beat.

B is wrong: Summation is prevented by a refractory period that lasts nearly as long as the twitch, and the stem fixes the stimulus interval in any case.

C is wrong: A larger trigger influx would raise the peak cytosolic Ca2+, and the stem states that the peak Ca2+ of each twitch is unchanged.

A competent response contains

· Ventricular sarcomeres stretched from 1.9 to 2.2 µm stay on the ascending limb, where overlap changes only modestly.

· Most of the extra force comes from length-dependent activation: stretch raises myofilament Ca2+ sensitivity and shifts the force versus Ca2+ curve to the left, so an unchanged Ca2+ transient recruits more cross-bridges.

Distinctions that matter

· Changing the Ca2+ transient and changing the sensitivity to it are separate levers, and the stem fixes the transient.

Common misconceptions

· Explaining the steep cardiac length-tension curve by overlap alone, which accounts for only part of the rise.

Where it is taught: Section C05 of the CV-5 Cardiac Muscle Contraction notes.